



CS 208: Mathematical Foundations of Computer Science

Course Code	CS 208
Course Name	Mathematical Foundations of Computer Science
L-T-P-C	3-1-0-4
Prerequisites	IC 150 Computation for Engineers
Intended for	UG
Distribution	Compulsory for CSE; CS elective for EE and ME
Semester	3rd
Approval	9th Senate (May 2016)

Course Contents

- **Fundamental structures:**
 - Functions - surjections, injections, inverses, composition. (2 contact hours) Relations - reflexivity, symmetry, transitivity, equivalence relations. [2 Lectures]
 - Sets - Venn diagrams, complements, Cartesian products, power sets, finite and infinite sets, introduction to lattices. [4 Lectures]
 - Abstract orders: quasi-order, partial order, well-order, (Advanced, optional topics: Zorn's lemma, Koenig's theorem.) [2 Lectures]
- **Combinatorics**
 - Counting arguments/techniques; pigeonhole principle; cardinality and countability, the inclusion-exclusion principle, recurrence relations, generating functions. [5 Lectures]
 - Basics of graph theory: graph as a discrete structure, graph coloring and connectivity, traversal problems, and spanning trees. [5 Lectures]
 - Advanced, optional topic: Probabilistic method in combinatorics.
- **Logic**
 - Propositional and predicate logic: syntax, semantics, soundness, completeness, unification, inferencing, resolution principle, proof system. [6 Lectures]
 - Proof techniques (negation, contradiction, contraposition, mathematical induction) and the structure of formal proofs; efficiency of proof-systems. [4 Lectures]
- **State machines**
 - Introduction, minimization, grammars, languages. [4 Lectures]
- **Algebra**
 - Motivation for algebraic structures, the theory of some algebras such as monoids, groups (finite, cyclic, permutation, matrix), cosets, subgroups, Lagrange's theorem, discrete logarithm. [8 Lectures]
- **Optional topic**
- **Number Theory**
 - Elementary number theory, fundamental theorem of arithmetic, gcd, unique factorization, Euler's function, modular arithmetic, Fermat's little theorem, Chinese remainder theorem, modular exponentiation, RSA public key encryption.

References

1. E. Lehman, F. T. Leighton, and A. R. Meyer, **Mathematics for Computer Science**, 2013. Available online at: <http://courses.csail.mit.edu/6.042/spring13/mcs.pdf>
2. R. L. Graham, D. E. Knuth, and O. Patashnik, **Concrete Mathematics**, Pearson, 1994. Also, available online at:
3. A. Aho and J. Ullman, **Foundations of Computer Science**, W. H. Freeman, 1992. Available online at: <http://infolab.stanford.edu/~ullman/focs.html>
4. I. N. Herstein, **Topics in Algebra**, 2nd Edition, Wiley, 1975.
5. A. Tucker, **Applied Combinatorics**, 6th Edition, Wiley, 2012.
6. C. Liu and D. P. Mohapatra, **Elements of Discrete Mathematics**, 3rd Edition, Tata-McGraw Hill, 2008.
7. T. Koshy, **Discrete Mathematics with Applications**, Academic Press, 2003.
8. J. Hein, **Discrete Structures, Logic, and Computability**, 3rd Edition, Jones and Barlett, 2009.